



UNIVERSITI TEKNOLOGI MALAYSIA

MID TERM TEST

SEMESTER 1, 2023/2024

SUBJECT CODE : SECJ 3553

SUBJECT : ARTIFICIAL INTELLIGENCE

TIME : 2 HOURS

DATE : 22 NOV 2023

INTRODUCTIONS TO STUDENTS:

This test book consists of 2 sections:

Part A: Multiple Choice Questions [15 Marks]

Part B: Theory and Applications [85 Marks]

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET.

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

Choose the correct answer for the following questions. Each question carries 1 mark.

1. What is the use of the Turing Test in the context of AI?
 - a. To evaluate a machine's ability to exhibit intelligent
 - b. Testing the speed of AI algorithms
 - c. Assessing the hardware capabilities of AI systems
 - d. Measuring the energy efficiency of AI models

2. What is knowledge representation in the context of artificial intelligence?
 - a. Storing information in databases
 - b. Encoding information in a form suitable for reasoning
 - c. Analyzing data for patterns
 - d. Transmitting information between agents

3. Which knowledge representation approach uses a graph-like structure to represent knowledge?
 - a. Rule-based systems
 - b. Semantic networks
 - c. Frames
 - d. Expert systems

4. Which of the following is not a propositional logic?
 - a. Johor Bahru is a country.
 - b. Kuala Lumpur is the capital of Indonesia.
 - c. Have a nice day everyone!
 - d. I am angry and hungry.

5. Consider the statements
P: Fara is rich.
Q: Fara is happy.

Then the symbolic form of the statement "Fara is poor or she is both rich and unhappy" is _____.
 - a. $\neg P \vee (P \wedge \neg Q)$
 - b. $P \vee (\neg P \wedge \neg Q)$
 - c. $\neg P \vee (\neg P \wedge Q)$
 - d. None of the above

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6. Which of the following represents the equivalent statement in clause form?

Statement: $\text{pass}(x, \text{exam}) \rightarrow (\text{holiday}(x) \wedge \text{shopping}(x))$

- a. $\neg \text{pass}(x, \text{exam}) \vee (\text{holiday}(x) \wedge \text{shopping}(x))$
- b. $\neg \text{pass}(x, \text{exam}) \wedge (\text{holiday}(x) \wedge \text{shopping}(x))$
- c. $\text{pass}(x, \text{exam}) \vee (\neg \text{holiday}(x) \vee \neg \text{shopping}(x))$
- d. $\text{pass}(x, \text{exam}) \wedge (\text{holiday}(x) \wedge \text{shopping}(x))$

7. Following are the steps in an inference technique named X

- i. Assume that our fact base is consistent.
- ii. Add the negation of the thing we want to prove to the fact base
- iii. Show that the fact base is now inconsistent
- iv. Conclude the thing we want to prove

What is X?

- a. Unification
- b. Modus Ponens
- c. Refutation Resolution
- d. Resolution

8. Based on the below premises,

$\forall x (\text{Person}(x) \wedge \text{HasFever}(x)) \rightarrow \text{MightHaveCOVID19}(x)$
 $\text{HasFever}(\text{John})$
 $\text{Person}(\text{John}) \wedge \text{HasFever}(\text{John})$

We can infer that John might have covid19 using:

- a. Modus Ponens
- b. And-Introduction
- c. Modus Tollen
- d. Universal Elimination

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9. In a course registration system, consider the following propositions:

A: "If a student has completed the prerequisite course, they can register for an advanced course."

B: "Tom has completed the prerequisite course."

C: "Tom wants to register for the advanced course."

Using proof methods, which of the following statements can be correctly derived based on the given propositions?

- a. Tom can register for the advanced course.
- b. Tom has registered for the advanced course.
- c. Tom has completed the advanced course.
- d. Tom wants to register for the prerequisite course.

10. In a course registration system, consider the following propositions:

A: "Tom has registered 14 credits"

B: "Tom has completed the prerequisite course."

C: "Tom has registered for the advanced course."

D: "Tom is in second semester."

Using And-Introduction, which of the following statements is **incorrectly** derived based on the given propositions?

- a. Tom is in second semester and Tom cannot register industrial training.
- b. Tom is in second semester and has registered for the advanced course.
- c. Tom is in second semester and has completed the prerequisite course.
- d. Tom is in second semester and has registered 14 credits.

11. Problem formulation is the process of deciding what actions and states to consider, given a goal. Problem formulated by specifying four things which are

- a. Admissible, monotonicity, completeness, complexity
- b. Completeness, optimality, time and space complexity
- c. Initial state, actions, goal test and path cost
- d. None of the above

12. Which of the following is the characteristic of Breadth-First Search?

- a. Inefficient when the goal is located at another branch.
- b. It is complete and optimal under most circumstances.
- c. It is either not complete or not optimal.
- d. It is not memory intensive and uses less space.

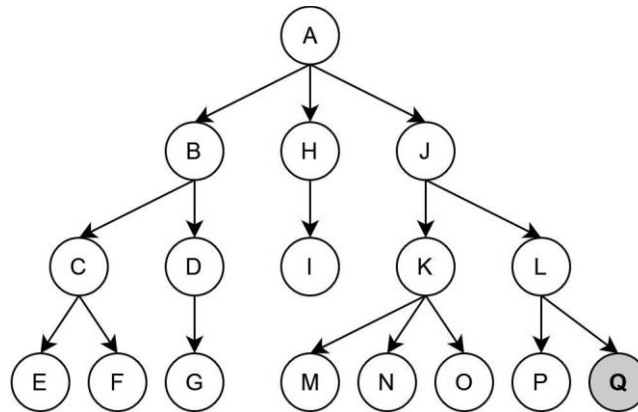


Figure 1

13. Based on the above state space graph, list the possible path using Breadth-First Search.
- A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q
 - A,B,C,E,F,D,G,H,I,J,K,M,N,O,L,P,Q
 - A,B,H,J,C,D,I,K,L,E,F,G,M,N,O,P,Q
 - A,B,H,J,C,E,F,D,G,I,K,M,N,O,L,P,Q
14. Which of the following scenarios illustrates the drawback of depth-first search in a graph?
- DFS always finds the shortest path between two nodes.
 - DFS may get stuck in a deep path before exploring shallower paths.
 - DFS guarantees optimal solutions in terms of path length.
 - DFS is not suitable for graphs with cycles.
15. What is a key difference between depth-first search (DFS) and breadth-first search (BFS) in a graph?
- DFS always explores shallower levels before deeper levels.
 - BFS uses a stack for traversal.
 - DFS explores a single branch as deeply as possible before backtracking.
 - BFS is always more memory-efficient than DFS.

SECTION B**TOTAL: 85 MARKS**

1. One aim of Responsible AI principles is to **Ensure ethical and fair use of AI technologies**. Match the six principles of Responsible AI with examples given.
(8 marks)

Example of Application	State the Related Principle of Responsible AI	Justification of the significance of your answer
A language translation tool that supports multiple languages and dialects, ensuring inclusivity for users worldwide		
a healthcare AI system that maintains a log of decision-making processes, allowing for easy auditing and accountability in patient diagnosis and treatment recommendations.		
a hiring platform that uses algorithms to ensure equal opportunities by avoiding gender or ethnicity biases in the selection process		
a virtual assistant platform may anonymize and encrypt user data, obtain explicit consent, and minimize data collection, ensuring privacy protection while delivering personalized services.		

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2. Translate the following English sentences into propositional logic using the symbolized key provided. (12 marks)

C	:	Canceled	TF	:	Tropical Feast
D	:	Delicious Street Food	SA	:	Savor
E	:	Enjoy The Meal	TH	:	Too Hot
F	:	Tropical Fruits In Season	RH	:	Raining Heavily
I	:	Indoor	M	:	Movies
R	:	Beautiful Reef	SC	:	Success
S	:	Sea	SN	:	Snorkeling
H	:	Happy			

- i. If there's delicious street food and they enjoy the meals, they are happy :
- ii. We visit tropical feast unless they're canceled or it's raining heavily :
- iii. We go snorkeling if and only if we're near the sea or at a beautiful reef :
- iv. If tropical fruits are in season, we savor them and have a tropical feast :
- v. We stay indoors and watch movies when it's too hot or raining heavily :
- vi. The party is success when the guest are happy and they enjoy the meals :

3. Given predicates below, convert the English sentences in parts (i) – (v) into standard predicate logic sentences. No (i) has already done for you. (10 marks)

student	ail_final_exam	participate_in_hackathons
enjoy_coding	graduate_with_honors	collaborate with classmates.
computer_science_major	excels_in_academics	participates_extracurricular_act

- i. *Some student are both enjoy coding and participates in hackathons*
- ii. Every student who is a computer science major and enjoys coding will either participate in hackathons or collaborate with classmates.

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4. Given the axioms:

- a. Ahmad is a hardworking student.
- b. All hardworking students go to the library.
- c. Students who go to the library will successfully pass the exam.

You are required to perform a refutation resolution to prove that *Ahmad will successfully pass the exam*.

Use the following notation to represent the problem.

S	:	Student
H	:	Hardworking
L	:	Go to Library
P	:	Pass the Exam

(10 marks)

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5. You are given the following scenario:

If Mary has coughing symptoms and shortness of breath symptom, then Mary is positive Covid19. If Mary has shortness of breath symptom, then Mary has coughing symptom. Mary has shortness of breath symptom.

Using the above scenario, construct a proof using inference rules for:
“Mary is positive COVID-19”

Use the following notation to represent the problem:

A: Mary has coughing symptom

B: Mary has shortness of breath symptom

C: Mary is positive COVID-19

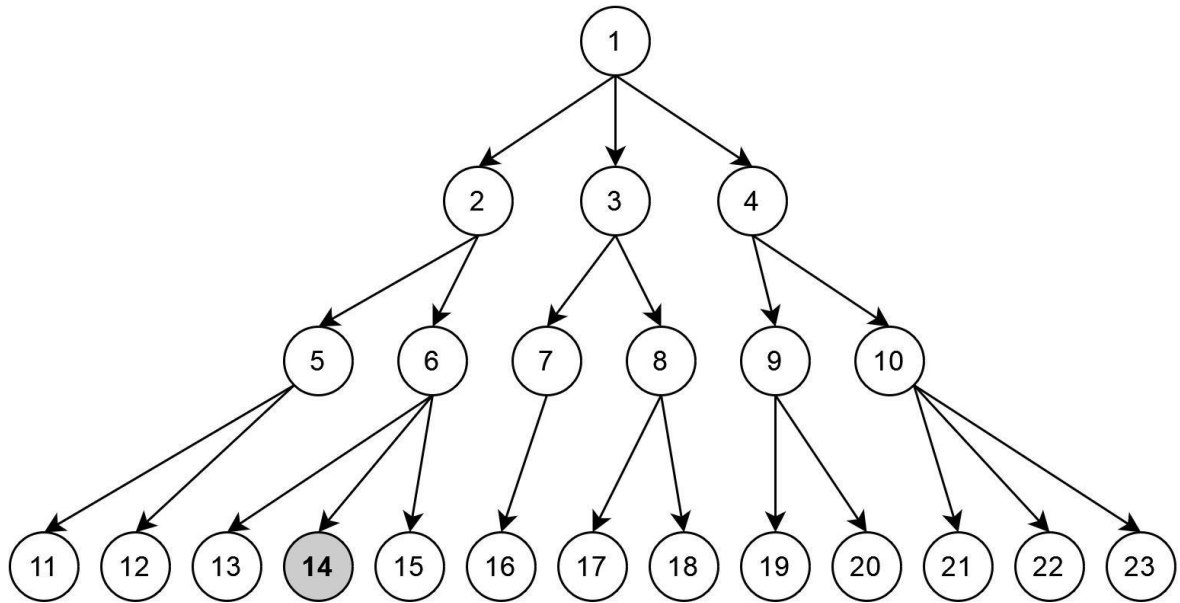


Figure 2.0

6. a. Perform Breadth-First Search on the state space graph in Figure 2.0 above. List down the order of nodes visited from starting node (Node 1) to the goal (Node 14) in a table as shown in Table 1.0.

** You may add more row if necessary.*

(13 marks)

Table 1.0

Iteration	Open List	Closed List
0	1	[]
1		

- a. What are the TWO advantages and ONE disadvantage of Breadth-First Search? Explain.

(3 marks)

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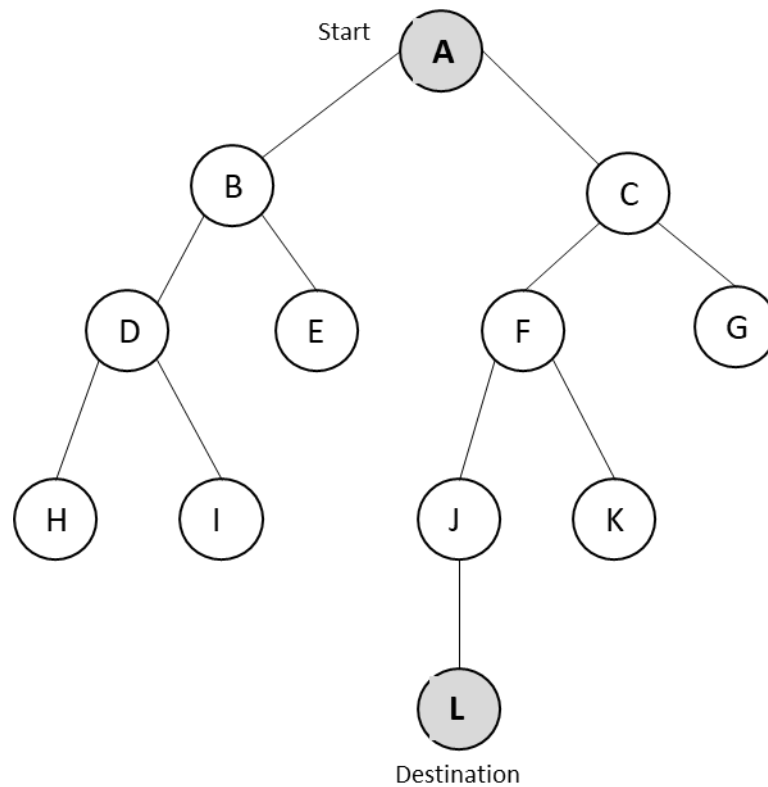


Figure 3.0

7. You are applying Depth-First Search (DFS) to find a path from node A to node L (see Figure 3.0). Initialize the open list, closed list, and start the traversal from node A

- a. Trace the search tree until you reach the destination (node L). Indicate the open list, closed list, and the current node at each step/iteration.

(15 MARKS)

- b. State the search path (from start to destination). (2 marks)